Ayenor Christopher

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Professional Summary

Electrical Engineering student passionate about turning complex ideas into real-world solutions. Combines hands-on expertise in embedded systems, PCB design, and hardware integration with a sharp ability to lead projects and solve problems creatively.

Education

Master's in Electrical and Computer Engineering (Meng) January 2025 – May 2027 (Expected)
University of New Brunswick, Fredericton, NB

Bachelor's in Information and Communication Engineering (BEng) September 2017 – August 2022
Covenant University, Ota, Ogun State

Skills

- **Hardware & Design Tools**: PCB Design (Altium Designer, JLCPCB), AutoCAD Electrical, Arduino, ESP32, STM32, Sensor Integration, FPGA, soldering, Quartus prime.
- **Electronics & Embedded Systems**: PCB troubleshooting, schematic design, microcontrollers, signal handling, real-time data processing.
- **Programming**: C, C#, Java, Python, Verilog.
- **Soft Skills**: Collaboration, problem-solving, detail oriented, adaptability, clear communication.
- **Collaboration Tools**: GitHub, Azure DevOps, Jira, Microsoft Teams.

Work Experience

❖ **Teaching Assistant – Digital Logic Design** Sept 2026 – Dec 2026
University of New Brunswick, Fredericton, New Brunswick

- Supported approximately 50 students across 8 laboratory sessions, providing hands-on guidance in digital circuit design, simulation, troubleshooting, and VHDL implementation.
- Marked and graded laboratory reports and quizzes for approximately 76 students, assessing their understanding of digital system design concepts.
- Evaluated VHDL-based projects, assessing code functionality, design correctness, and implementation requirements.
- Assisted students in troubleshooting VHDL code and resolving digital circuit design and implementation challenges.
- Provided technical guidance on digital logic, hardware design principles, and VHDL programming during laboratory sessions.

❖ **Electrical Engineer (Intern)** Jan 2026 – Present
SmartSkin Technologies, Fredericton, New Brunswick

- Diagnosed and repaired 30+ PCB assemblies at component level, identifying hardware faults through systematic testing and inspection.
- Recommended hardware design improvements to simplify troubleshooting and enhance assembly efficiency.
- Performed diagnostics and troubleshooting on industrial electronic systems using multimeters and other electronic test equipment.
- Conducted root cause analysis and implemented corrective actions to address recurring hardware failures and improve product reliability.
- Supported electronics quality assurance through systematic inspection, testing, and documentation of repair activities.

❖ **Software and Hardware Engineer Intern** January 2023 – December 2024
Precise Financial Systems, Lagos, Nigeria

- Performed diagnostic testing and component replacement on personal computers and electronic equipment, reducing device downtime by 30%.
- Diagnosed and repaired over 3 laptops and electronic devices, resolving hardware and software faults to restore full functionality.

- Engineered full-stack features using C# (.NET 7) for backend services and React / Next.js for modern, responsive user interfaces, delivering high-performance solutions for financial and logistics operations.
- Participated in peer reviews and maintained high code quality standards, contributing to a 90% reduction in post-release bugs across both applications.

Projects

❖ **STM32 PCB Design with Altium** September 2025 – December 2025

- **Tech Used:** Language: Tools: Altium Designer | Hardware: STM32F4 MCU, USB interface, Sensor, GPIO header, Power supply circuitry.
- **Impact:** Designed a four-layer STM32F4 PCB with USB, sensor integration, GPIO expansion, and power regulation for a compact embedded platform.
- **Skills Gained:** PCB schematic design, multi-layer layout and routing, high-speed USB signal handling, power supply design, and hardware validation.
- **Achievements:** Completed full end-to-end design process from schematic to Gerber generation.

❖ **Hand Tracking with Arduino Servo Control** September 2025 – December 2025

- **Tech Used:** Language: Python, C | Tools: OpenCV, cvzone, pyFirmata, GitHub | Hardware: Arduino Uno, Servo Motor, Webcam
- **Impact:** Built a gesture-controlled physical interface that translates real-time hand tracking into servo motor movement, enhancing understanding of computer vision, embedded systems, and human-computer interaction.
- **Skills Gained:** Computer vision with OpenCV, hand landmark detection with cvzone, serial communication with Arduino, servo control using Python, real-time signal processing, hardware-software integration.
- **Achievements:** Designed and implemented a real-time gesture recognition system that counts raised fingers and moves a servo accordingly.

❖ **Remote-Based Battery Monitoring with ESP32S** September 2025 – December 2025

- **Tech Used:** Language: C | Tools: Arduino IDE, Blynk | Hardware: ESP32, Voltage/Current Sensors
- **Impact:** Developed a wireless system that monitors battery voltage and current in real time and transmits data remotely for improved energy management.
- **Skills Gained:** IoT communication (Wi-Fi), real-time data acquisition, and soldering of electrical components.
- **Achievements:** Successfully implemented a low-cost monitoring solution capable of streaming battery performance metrics remotely with high accuracy.

School Experience

❖ **Advanced Digital Systems (EE-6213)** September 2025 – December 2025

University of New Brunswick Fredericton, Fredericton, NB

Topics Included:

- Design methods and tools for FPGA-based digital circuits focusing on large-scale systems
- Simulation, testing, and verification techniques for FPGA circuits

Outcomes:

- Currently developing practical skills in FPGA design, implementation, and optimization through real-world projects.
- Gaining experience with simulation and verification tools to ensure design correctness and performance.

Learnings:

- Understanding of the full VLSI design lifecycle tailored to FPGA architecture.

Certifications

- Microcontroller Embedded C Programming (Udemy)
- Engineering project management: scope, time and cost management (Coursera)
- WHMIS (Workplace Hazardous Materials Information System)
- Introduction to Electronics (Coursera)
- Embedded System (Alison)
- Programming for everybody (Python) (Coursera)